

ERP ↔ Thaisausage Integration Test Plan

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Audience: ERP team, Thaisausage IT/Operations and eVRP test coordinator
System under test: <https://thaisausage.krs.co.th>

1. Test objective

ยืนยันการรับ-ส่งข้อมูลระหว่าง ERP และ thaisausage ตั้งแต่การแจ้งเหตุการณ์ SO การอ่าน SO จาก SQL Server แบบ SELECT-only การส่ง standard JSON ไปยัง eVRP การรับ DO เข้าพื้นที่ staging และการตอบกลับสถานะให้ผู้เรียก โดยไม่เปลี่ยนข้อมูลจริงใน ERP และไม่ส่งข้อมูลจริงไป eVRP จนกว่าจะผ่าน UAT และได้รับอนุมัติ

2. Scope and direction

Direction	Flow	Current scope	Expected side effect
ERP → Thaisausage	direct SO / DO receipt; legacy webhook retained	Yes	SQLite submission/staging only
Thaisausage → ERP	HTTP response to ERP caller	Yes	response only; no ERP write-back
ERP SQL → Thaisausage	approved SO SELECT	Yes	SELECT-only
Thaisausage → eVRP	mapped SO submission	scheduled worker/mock first	no real eVRP request in dry-run
eVRP → Thaisausage	DO callback to <code>/api/v1/vrp/do-received</code>	Yes	SQLite staging only; never writes ERP
eVRP → Thaisausage	response/status	mock/UAT	submission state and reference
Thaisausage → ERP	asynchronous callback	Not available	separate future contract

3. Safety rules

1. Local tests use temporary SQLite and mocked ERP/eVRP connectors. 2. Deployed smoke uses `dry_run=true` and `sync.enabled=false`. 3. Real ERP SQL Server may be queried only for SO numbers explicitly approved by the user. 4. SQL must be the reviewed SELECT query with a read-only account. No INSERT, UPDATE, DELETE, MERGE, EXEC, DDL or ERP status update. 5. DO tests use mock payloads and must prove `erp_write=false`. 6. No real eVRP submission before the UAT window, test token and sign-off are approved. 7. Never log API keys, SQL passwords, full customer data or unredacted payloads.

4. Entry criteria

- [] `HTTPS`, `/health`, `/docs` and `/openapi.json` return successfully.
- [] ERP has received the OpenAPI contract and has a test sender.
- [] Test API key is configured through secret storage.

- [] `dry_run=true` and `sync.enabled=false` are confirmed.
- [] Approved SQL query, field mapping and approved SO list are recorded.
- [] Test owner, observer and rollback contact are available.
- [] eVRP test URL/token and test master data are available before eVRP UAT.
- [] Scheduled worker is the only SO trigger; `sync.interval_seconds` is agreed for the test window.
- [] `sqlserver.approved_orders_query` is reviewed and filters `IsApprSo = 1`; the app refuses a query without that predicate.
- [] `do_write.enabled=false` is confirmed on the environment under test.

5. ERP → Thaisausage test cases

ID	Scenario	Action	Expected result	Evidence
ERP-01	Health/contract	GET <code>/health</code> , <code>/docs</code> , <code>/openapi.json</code>	HTTP 200; no secret exposed	redacted response
ERP-02	Valid SO hook	POST <code>/api/v1/erp/hooks/order-ready</code> with unique <code>event_id</code>	202, <code>state=received</code> , <code>sweep=queued</code>	<code>event_id/status</code>
ERP-03	Hook replay	resend identical body	idempotent; one logical event	hook count/status
ERP-04	Hook conflict	same <code>event_id</code> , changed order data	409 or safe conflict; no second action	response
ERP-05	Invalid hook	missing ID/unknown event type	422; not processed	response
ERP-06	Valid direct SO	POST <code>/api/v1/erp/orders</code> with standard JSON	200/202; request stored	<code>request_id/status</code>
ERP-07	SO replay	same <code>request_id</code> and identical payload	idempotent; no duplicate send	submission lookup
ERP-08	Changed payload	same <code>request_id</code> with changed line	409; operator review	response
ERP-09	Validation	bad date, quantity, hub or duplicate order	422; no outbound attempt	response
ERP-10	DO staging	POST <code>/api/v1/erp/do-received</code> mock receipt	202, <code>state=staged</code> , <code>erp_write=false</code>	<code>receipt/status</code>
ERP-11	DO replay/conflict	same receipt, then changed payload	replay safe; changed payload 409	receipt lookup
ERP-12	Authentication	no/wrong/correct Bearer key	401/401/accepted	status only
ERP-13	Network retry	timeout/disconnect then retry same body	same identity; reconcile review state	IDs/state
ERP-14	Scheduled approved SO	run the schedule with a reviewed query	only approved SOs are selected; no Hook required	timestamps/query result
ERP-15	Unapproved SO excluded	include an SO with <code>IsApprSo <= 1</code> in the source data	the SO is never selected and never reaches eVRP	query result/submission list
ERP-16	Query guard	configure a query without the <code>IsApprSo</code> predicate	startup/sweep refuses with a contract error; no SQL executed	error message
ERP-17	Sync disabled	set <code>sync.enabled=false</code> and wait two intervals	no SQL query runs and no SO is sent	worker log/query trace
ERP-18	Dry-run schedule	<code>dry_run=true</code> with an approved SO	result <code>preview</code> ; no eVRP call and no <code>order_claims</code> row	state/database
ERP-19	Duplicate protection	run two cycles over the same approved SO	second cycle reports <code>skipped / already_sent</code>	submission/claim rows
ERP-21	Unresolved attempt	force <code>needs_review</code> , then run another cycle	second cycle reports <code>needs_review / prior_attempt_needs_review</code> and does not resend	states/upstream calls

ID	Scenario	Action	Expected result	Evidence
ERP-22	Incomplete detail	approved SO whose detail row has no item code	SO becomes <code>review / detail_item_code_missing</code> and is not sent	per-order result
ERP-23	Unreadable batch	query returns a row without the configured order key	cycle returns one <code>review</code> entry, no exception, log shows counts only	worker log
ERP-20	Isolation	include one unmappable SO next to a valid one	bad SO becomes <code>review</code> ; the valid SO is still sent	per-order results

5.1 eVRP → Thaisausage DO callback test cases

eVRP เรียก `POST /api/v1/vrp/do-received` หลังสร้าง DO เสร็จ ทุกเคสใช้ข้อมูลจำลองเท่านั้น

ID	Scenario	Action	Expected result	Evidence
DO-01	Valid callback	POST with unique <code>receipt_id</code> and <code>do_no</code>	202, <code>state=staged</code> , <code>source=eVRP</code> , <code>receipt_key=eVRP:<receipt_id></code> , <code>erp_write=false</code>	receipt/status
DO-02	Replay	resend the identical payload	202 with <code>replayed=true</code> ; one row in <code>do_receipts</code>	row count
DO-03	Conflict	same <code>receipt_id</code> , changed payload	409; original staged row unchanged	response/row
DO-04	Unauthorized	omit or corrupt the Bearer key	401; nothing stored	status only
DO-05	Invalid JSON	malformed body or wrong Content-Type	422; nothing stored	status only
DO-06	Missing identity	body without <code>receipt_id</code> and <code>do_no</code>	422; nothing stored	status only
DO-07	No ERP write	inspect ERP after DO-01 to DO-06	zero ERP rows created/changed; <code>do_write.enabled=false</code>	before/after counts
DO-08	No echo to eVRP	observe outbound traffic during DO tests	Thaisausage sends nothing back to eVRP	redacted network log
DO-09	Concurrent delivery	send the same callback twice at the same moment	one staged row; the other answer is <code>replayed=true</code> ; never HTTP 500	row count/status
DO-10	Source isolation	same <code>receipt_id</code> from ERP path and eVRP path	two separate staged rows, no conflict	<code>receipt_key</code> values
DO-11	Writer disabled	callback while <code>do_write.enabled=false</code>	202, <code>erp_write=false</code> , <code>do_write.state=disabled</code> , no ERP row	response/ERP counts
DO-12	No transaction_no	callback without <code>transaction_no</code>	202, <code>do_write.state=skipped</code> , DO still staged	response/staging row
DO-13	Writer rejects	enabled writer with incomplete mapping	202, <code>do_write.state=rejected</code> , no SQL executed	response/log
DO-14	Duplicate transaction	second DO reusing a written <code>transaction_no</code>	202, <code>do_write.state=conflict</code> , one ERP document only	response/ERP counts

6. Thaisausage → ERP SQL Server test cases

ID	Scenario	Action	Expected result	Evidence
SQL-01	Connection	use connection string plus ERP_SQL_USER/PASSWORD	connects; credentials not logged	redacted check
SQL-02	Approved SO	run reviewed query for one approved SO	rows read and grouped correctly	row count/order
SQL-03	Unknown SO	query test/nonexistent key	empty/controlled review; no broad sweep	result count
SQL-04	Mutation guard	try mutation/non-SELECT query	rejected before execution	rejection code
SQL-05	Mapping	COD, credit, free item, Thai text, decimals, multi-line	standard JSON matches contract	redacted JSON
SQL-06	Scheduled sweep	run approved-SO schedule in controlled mode	correct submission preview/state; no ERP write	run/result state
SQL-07	SQL failure	simulate timeout/unavailable server	controlled review; scheduler stays healthy	redacted logs
SQL-08	Approved schedule	run configured approved-orders query with IsApprSo = 1	only approved SOs are returned; no Hook required	query and result count

7. Thaisausage → eVRP and response tests

ทดสอบด้วย mock eVRP ก่อนเสมอ และใช้ eVRP UAT เท่านั้นเมื่อได้รับอนุมัติ

ID	Scenario	Action	Expected result	Evidence
VRP-01	Valid outbound	submit mapped mock SO	token header reaches mock only	redacted request
VRP-02	Accepted	mock HTTP 200 with reference	state <code>sent</code> ; reference stored	submission
VRP-03	Temporary failure	mock 502/timeout	<code>needs_review</code> ; no automatic resend; operator reconciliation	redacted response
VRP-04	Ambiguous timeout	accept then close connection	no blind changed-payload resend	review item
VRP-05	Business reject	mock 422	not marked sent; actionable error	response
VRP-06	Duplicate	repeat identical request	idempotent logical result	ID/reference
VRP-07	Auth failure	mock 401/403	no uncontrolled loop; alert operator	status
VRP-08	Unknown hub	send a hub code absent from the eVRP master	422 <code>PICKUP_HUB_NOT_FOUND</code> ; message stored in <code>upstream_error</code>	submission record
VRP-09	Customer without master	send an SO whose customer has no delivery point/route	422 <code>CUSTOMER_MASTER_REQUIRED</code> ; no retry	submission record
VRP-10	Identity reuse	resend a corrected order under the previous request_id	409 <code>REQUEST_ID_CONFLICT</code> ; the scheduler must generate a new id instead	request_id values

ERP must treat 202 as accepted for processing, not proof of eVRP completion. ERP may retry transport failures with the same identity. Thaisausage does not run an automatic retry worker or attempt history yet. A 409 stops automation. A 422 requires payload correction. `needs_review` requires operator reconciliation.

8. Test execution record

For every case record: test run ID, environment, Bangkok time, tester, endpoint, HTTP status, request/event/receipt ID, expected state, actual state and redacted evidence path.

At the end record:

- ERP rows before = ____; ERP rows after = ____; mutation check = pass/fail
- Real eVRP send = not used / approved UAT
- Passed = ____; failed = ____; blocked = ____
- Defects, owner, corrective action and retest result

9. Exit gates and sign-off

Gate A — technical smoke

- ☐ ERP-01 to ERP-05 pass.
- ☐ HTTPS, authentication and CSP/Swagger are verified.
- ☐ No secret appears in evidence.

Gate B — controlled data flow

- ☐ ERP-06 to ERP-20 pass with mock/staging data.
- ☐ DO-01 to DO-08 pass with mock callbacks.
- ☐ SQL-01 to SQL-07 pass using only approved SO SELECTs.
- ☐ DO receiver proves `erp_write=false`.
- ☐ ERP row/status counts are unchanged.

Gate C — eVRP UAT

- ☐ VRP-01 to VRP-07 pass against eVRP test environment.
- ☐ eVRP confirms customer, item, hub and remote reference mapping.
- ☐ Retry, duplicate and recovery behavior is accepted.
- ☐ Production token, live mode and schedule are approved separately.

Role	Name	Result	Date/signature
ERP owner		Pass / Fail	
Thaisausage IT		Pass / Fail	
eVRP coordinator		Pass / Fail	
Business/data owner		Pass / Fail	

10. Related contracts

- Swagger: <https://thaisausage.krs.co.th/docs>
- OpenAPI: `docs/openapi.json`
- ERP guide: `docs/ERP_Integration_Guide.md`
- Operations guide: `docs/Thaisausage_Operations_Guide.md`